

Clinical Risk Assessment Report: Patient 38

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 9.8% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.44x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Patient Age (Age), which reduced the patient's absolute risk by 1.5%. Biologically, this feature patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Patient Age (Age) **Value: 63.0 [Avg (47th %ile)] (-1.5%)**

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.42 [Avg (41th %ile)] (-1.3%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: 0.27 [Top 13%] (-1.2%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Carcinoembryonic Antigen (CEA) **Value: 2.0 [Avg (26th %ile)] (-1.0%)**

Elevated levels are strongly associated with increased tumor burden and potential for micrometastasis.

Tumor Local Contrast (PET) (GLCM_Contrast(=Variance).PET) **Value: -0.52 [Avg (35th %ile)] (-0.7%)**

Measures local variations and sharp transitions in density or metabolism, signifying an unstructured and variable tumor matrix.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: 0.1 [Top 13%] (-0.6%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Minimum Tumor Density (CONVENTIONAL_HUmin) **Value: -1.15 [Bottom 11%] (-0.6%)**

Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: -1.68 [Bottom 8%] (-0.5%)**

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Machine Learning Feature Attribution (SHAP Decision Path)

